

Fall 2009

Problem 1 (Fa09). Let $p(k)$ be a degree n polynomial with complex coefficients defined by

$$p(k) = p_0 + \binom{k}{1}p_1 + \binom{k}{2}p_2 + \cdots + \binom{k}{n}p_n$$

where $p_0, \dots, p_n$ are complex constants. Define

$$f(z) = \sum_{k=0}^{\infty} p(k)z^k$$

Find the radius of convergence of the power series, prove that $f(z)$ is a rational function restricted to the disc of convergence, and give a formula for $f(z)$.

Problem 2 (Fa09). Prove that no polynomial $p(a, b, c, d)$ in four variables over $\mathbb{C}$ has the property that when p is evaluated on the entries of a 2×2 matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ the result is an eigenvalue of A , for all A .

Problem 3 (Fa02, Fa09). Evaluate the integrals

$$I_n = \int_{C_n} \frac{1}{z^3 \sin z} dz \quad n = 0, 1, \dots$$

where C_n is the circle $|z| = (n + \frac{1}{2})\pi$, with the counterclockwise orientation.

Problem 4 (Fa09). Let p, n be positive integers with p prime. Let $G = \text{GL}_n(\mathbb{Z}_p)$ be the group of invertible $n \times n$ matrices over the field with p elements. Let $U \subset G$ be the subgroup consisting of upper triangular matrices with all diagonal entries equal to 1. Prove that every p -subgroup of G is conjugate to a subgroup of U .

Problem 5 (Fa09). Is it possible to find two closed and connected subsets A and B in $\mathbb{R}^2$ such that

$$A + B = \{(x_1 + x_2, y_1 + y_2) \in \mathbb{R}^2 \mid (x_1, y_1) \in A, (x_2, y_2) \in B\}$$

is not closed?

Problem 6 (Fa09). For $n \geq 1$, prove that

$$a_n + \frac{1}{a_{n-1} + \frac{1}{\ddots + \frac{1}{a_1 + \frac{1}{a_0}}}} = \frac{\Delta_n}{\Delta_{n-1}}$$

where

$$\Delta_n = \begin{vmatrix} a_0 & 1 & 0 & & 0 \\ -1 & a_1 & 1 & & \\ 0 & -1 & a_2 & & \\ & & & \ddots & \\ 0 & & & 0 & -1 & a_n \end{vmatrix}$$

Problem 7 (Fa09). If f is a meromorphic function which has a pole of order 1 at z_0 show that

$$\frac{f'''(z)}{f'(z)} - \frac{3}{2} \left(\frac{f''(z)}{f'(z)} \right)^2$$

can be extended to a holomorphic function at z_0 .

Problem 8 (Fa09). Find the number of elements a in the ring $R = \mathbb{Z}_{6464}$ such that $a = x^5$ for some $x \in R$

Problem 9 (Fa09). Show that

$$\lim_{n \rightarrow \infty} \left(1 + \frac{z}{n} \right)^n = e^z$$

uniformly on compact subsets of $\mathbb{C}$.

Problem 10 (Fa09). Find the real values of β for which the following limit exists and is finite:

$$\lim_{R \rightarrow \infty} \int_1^R x^2 \cos x^\beta \, dx$$

Problem 11 (Fa09). Let P be a square matrix over $\mathbb{R}$ such that $P^\top P = P$. Prove that there exists a matrix A such that $A^\top A$ is invertible and $P = A(A^\top A)^{-1} A^\top$.

Problem 12 (Fa09). Prove that the function

$$f(z) = \frac{z}{(1-z)^2}$$

is injective on the disc $\mathbb{D}$. Find its Maclaurin series and determine its radius of convergence. What is the maximal disc $B_r(0) = \{w \in \mathbb{C} \mid |w| < r\}$ such that $B_r(0) \subset f(\mathbb{D})$?

Problem 13 (Fa09). Determine the number of rotationally distinct colorings by two colors of the edges of a regular tetrahedron T .

Problem 14 (Fa09). Solve the differential equation $x \frac{dy}{dx} + y = y^2$ for $y(1) = 2$.

Problem 15 (Fa09). Let $\langle z, w \rangle = \sum z_i \bar{w}_i$ be the Hermitian inner product on $\mathbb{C}^n$, and let A be a *normal* linear operator on $\mathbb{C}^n$, i.e., $A^*A = AA^*$ where $A^* = (\bar{A})^T$ is Hermitian adjoint to A . Prove that the set of complex numbers

$$\Lambda_A = \{\langle Az, z \rangle \mid z \in \mathbb{C}^n, \langle z, z \rangle = 1\}$$

is a convex polygon.

Problem 16 (Fa78, Fa09). Evaluate $\int_0^{2\pi} \frac{d\theta}{1 - 2r \cos \theta + r^2}$ where $|r| \neq 1$.

Problem 17 (Fa09). Let n be a positive integer and $\mathbb{Z}_n$ the cyclic group of order n and S_n the symmetric group on n letters. For each of the groups

$$\mathbb{R}^* \quad \mathbb{C}^* \quad \mathbb{Z}_2 \times \mathbb{Z}_3 \quad S_4 \quad \text{SL}_2(\mathbb{R})$$

prove or disprove that $\mathbb{Z}_6$ is isomorphic to a subgroup of it.

Problem 18 (Fa88, Fa09). If f is a smooth real valued function of x, y, z such that

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} > 0$$

for all x, y, z , show that f does not have a local maximum.